

Contract Reference: DGT/ISA 3044-C
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Deliverable 3 Report

Deliverable 3 was on creating the final product that can be used to run predictions across all of Portugal. My solution was to set up a pipeline in GitHub, where only one submission needs to be made in order to process as many HDF5 files as needed. The GitHub repository has been copied to the CNCA server, which is where it will usually be ran due to the large HDF5 file sizes.

The pipeline can start with either an individual HDF5 file containing Sentinel-2 data for the tile or with a directory of HDF5 Sentinel-2 files. From there, it will read the HDF5 files, determine change dates to use, split the file into blocks (6x6 cubes of 256x256 pixel chips), and distribute blocks to different CPUs. The individual CPUs handle creating the time composites and making the predictions for all the chips in the block. Once those predictions are done, one CPU reads all of the results from all of the blocks and stitches them together to create one overall tile output. If a directory of files was inputted, there is an additional step of stitching all of the tile prediction results together to create a full country output.

The pipeline has many variables that can be adjusted as needed by the user. The majority of these are optional and are set to a default that works best with our current use, but there is documentation in the code base explaining what the variables change and how to implement them. The pipeline can also be ran with different earth observation models, so theoretically DGT or other government organizations can use the same setup with their own model and not have to create a whole new process of downloading, storing, and cleaning Sentinel-2 data.

The code for this deliverable can be found on the GitHub project at the link below.

Project GitHub link - https://github.com/S2change/vegetation_loss